



Tianwen-1

All the details about Tianwen-1, the ambitious plan by China to put a rover on the surface of Mars. <https://digit.in/jul20-79>



Exo Mars rover

The instruments aboard ESA's Rosalind Franklin rover which would have used this window. <https://digit.in/jul20-80>

mission in the sense that a drill will dig up core samples of the regolith and then stash it on the surface for retrieval and study by future missions. Apart from the astro-biology goals, the rover will also explore the geology of the surface, and demonstrate the capabilities of producing oxygen in situ on the surface of Mars for future missions by human pioneers.

On board the rover are seven scientific payloads. Just like Curiosity, Perseverance also has a Mastcam, the new and improved Mastcam-Z. This camera is capable of stereoscopic imaging, zoom and can capture panoramas of the terrain. Then there is the SuperCam, which combines chemical analysis and imaging technologies, so that the mineral composition of rocks can be determined even at a distance. The Planetary Instrument for X-ray Lithochemistry (PIXL) instrument is an X-ray fluorescence spectrometer that can analyse the chemical composition of its study subjects in incredible detail. Then there is the Scanning Habitable Environments with Raman & Luminescence for Organics and Chemicals (SHER-LOC), which is the first instrument of its kind to be deployed on Mars. This instrument also includes a high resolution colour camera, and will be used to characterize minerals as well as organic compounds. The Mars Environmental Dynamics Analyzer (MEDA) is a tiny weather station that can monitor the temperature, wind speed and humidity on Mars. The Radar Imager for Mars' Subsurface Experiment (RIMFAX) is a ground penetrating radar that will be used to image the subsurface on Mars. Finally, there is The Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE), that will be used to demonstrate the conversion of atmospheric carbon dioxide into oxygen.

The rover will have a very special payload, which is the Ingenuity helicopter. The blades will rotate about

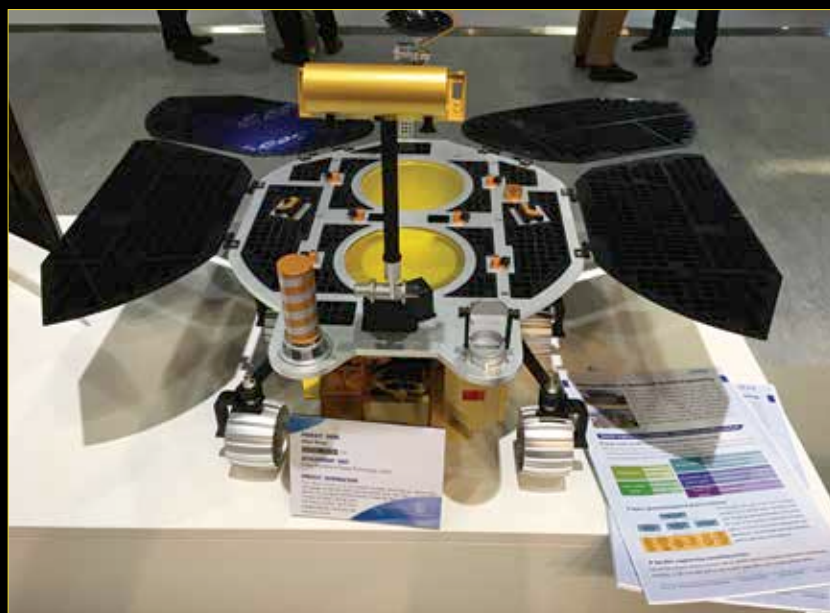
8 times as fast as the helicopters on Earth to produce enough lift on the tenuous atmosphere. Ingenuity can be used to explore areas that would otherwise be inaccessible to the rover. If all goes according to plan, it will be the first aircraft to perform a controlled flight on another planet. If successful, a helicopter can become a regular component of future NASA missions to Mars.

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Tianwen-1 is a Chinese mission to Mars that is made up of a rover, a lander and an orbiter. It is scheduled for a late July launch on

spectrometer, an orbital subsurface radar and the Matter and Ion Neutral Particle Analyser (MINPA). All of these instruments are for investigating the geology of the planet, and exploring the surface as well as the subsurface.

On board the rover is a ground penetrating radar, which is something that Perseverance has as well. This instrument will be capable of exploring the subsurface upto a depth of 100 meters. There is an instrument to measure magnetic fields on the surface, and another one to analyse chemical compounds in the environment.



Mockup of the Tianwen Rover

board a Long March 5 rocket. The mission objectives are very similar to the Mars 2020 mission, and Tianwen-1 also hopes to look for signs of past life on Mars, as well as characterize the geology of the planet. The orbiter and the rover will both be loaded with a number of scientific payloads.

On the orbiter, are a pair of high resolution and medium resolution cameras which can image the surface at 2 meter and 100 meter resolutions from an altitude of 400 kilometers. Then there is a Mars magnetometer, a Mars mineral

Like Perseverance, Tianwen-1 will also have a tiny weather station on board, which can characterize the local environment on the surface. The rover will be equipped with two cameras, a multi spectrum camera for imaging the surface, and a navigation and topography camera. The rover will look for signs of past (and present) life by looking for biosignatures and characterise the Martian regolith and investigate the distribution of water ice on the surface. The orbiter will study the atmosphere, as well as the Martian ionosphere. 